ASSIGNMENT-9

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Consider the following tables namely "DEPARTMENTS" and "EMPLOYEES" Their schemas are as follows,
Departments (dept no , dept name , dept_location); Employees (emp_id , emp_name , emp_salary,dept_no);

a) Develop a query to grant all privileges of employees table into departments table
b) Develop a query to grant some privileges of employees table into departments table
c) Develop a query to revoke all privileges of employees table from departments table
d) Develop a query to revoke some privileges of employees table from departments table
e) Write a query to implement the save point.

CODE:

```
CREATE DATABASE IF NOT EXISTS college;
USE college;

SET SQL_SAFE_UPDATES = 0;

DROP TABLE IF EXISTS Employees;
DROP TABLE IF EXISTS Departments;


CREATE TABLE Departments (
   dept_no      INT PRIMARY KEY,
   dept_name     VARCHAR(50),
   dept_location VARCHAR(50)
);

CREATE TABLE Employees (
   emp_id     INT PRIMARY KEY,
   emp_name   VARCHAR(50),
   emp_salary DECIMAL(10, 2),
   dept_no    INT,
   FOREIGN KEY (dept_no) REFERENCES Departments(dept_no)
);
```

```
INSERT INTO Departments VALUES
(10, 'HR', 'Delhi'),
(20, 'Finance', 'Mumbai'),
(30, 'Mathematics', 'Pune'),
(40, 'Commerce', 'Chennai'),
(50, 'Science', 'Kolkata');

INSERT INTO Employees VALUES
(1, 'Gaurav', 60000, 10),
(2, 'Sneha', 75000, 20),
(3, 'Amit', 55000, 30),
(4, 'Priya', 80000, 40),
(5, 'Rahul', 70000, 50);
```

Result Grid | Filter Rows: | Edit

emp_id	emp_name	emp_salary	dept_no
1	Gaurav	60000.00	10
2	Sneha	75000.00	20
3	Amit	55000.00	30
4	Priya	80000.00	40
5	Rahul	70000.00	50
NULL	NULL	NULL	NULL

```
SELECT * FROM Employees;

DROP USER IF EXISTS 'dept_user'@'localhost';
CREATE USER 'dept_user'@'localhost' IDENTIFIED BY 'password123';

GRANT ALL PRIVILEGES ON college.Employees TO 'dept_user'@'localhost';
SHOW GRANTS FOR 'dept_user'@'localhost';
```

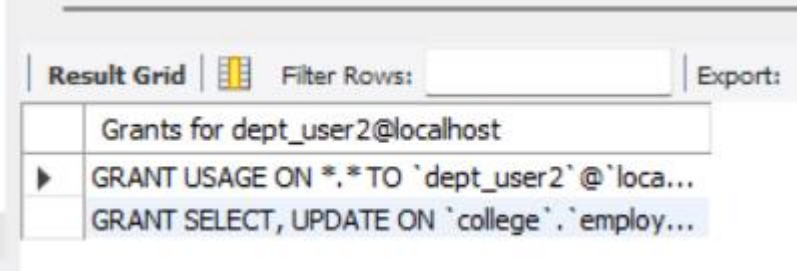
Result Grid | Filter Rows: | Export:

Grants for dept_user@localhost
GRANT USAGE ON *.* TO `dept_user`@`local...
GRANT ALL PRIVILEGES ON `college`.`employe...

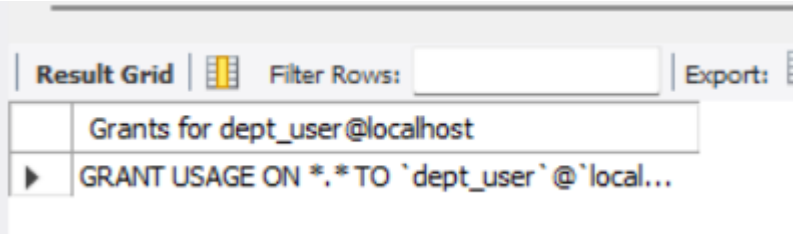
DROP USER IF EXISTS 'dept_user2'@'localhost';
CREATE USER 'dept_user2'@'localhost' IDENTIFIED BY 'password456';

GRANT SELECT, UPDATE ON college.Employees TO 'dept_user2'@'localhost';

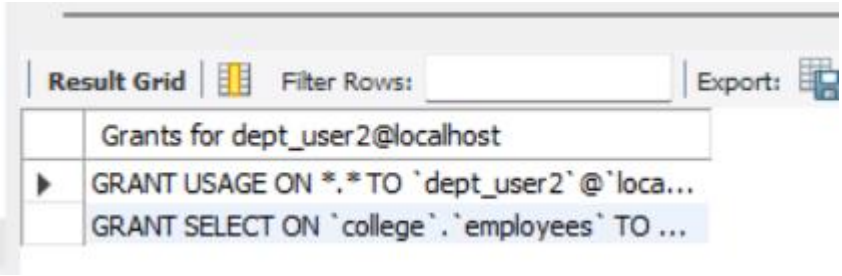
SHOW GRANTS FOR 'dept_user2'@'localhost';



REVOKE ALL PRIVILEGES ON college.Employees FROM 'dept_user'@'localhost';
SHOW GRANTS FOR 'dept_user'@'localhost';



REVOKE UPDATE ON college.Employees FROM 'dept_user2'@'localhost';
SHOW GRANTS FOR 'dept_user2'@'localhost';



START TRANSACTION;

INSERT INTO Employees VALUES (6, 'Kiran', 65000, 10);
INSERT INTO Employees VALUES (7, 'Meena', 72000, 20);

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Gaurav	60000.00	10
2	Sneha	75000.00	20
3	Amit	55000.00	30
4	Priya	80000.00	40
5	Rahul	70000.00	50
6	Kiran	80000.00	10
7	Meena	90000.00	20
NULL	NULL	NULL	NULL

SAVEPOINT sp1;

UPDATE Employees SET emp_salary = 80000 WHERE emp_id = 6;

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Gaurav	60000.00	10
2	Sneha	75000.00	20
3	Amit	55000.00	30
4	Priya	80000.00	40
5	Rahul	70000.00	50
6	Kiran	80000.00	10
7	Meena	72000.00	20
NULL	NULL	NULL	NULL

SAVEPOINT sp2;

UPDATE Employees SET emp_salary = 90000 WHERE emp_id = 7;

SELECT * FROM Employees;

ROLLBACK TO sp2;
SELECT * FROM Employees;

Result Grid | Filter Rows: | Edit:

emp_id	emp_name	emp_salary	dept_no
1	Gaurav	60000.00	10
2	Sneha	75000.00	20
3	Amit	55000.00	30
4	Priya	80000.00	40
5	Rahul	70000.00	50
6	Kiran	65000.00	10
7	Meena	72000.00	20
NULL	NULL	NULL	NULL

ROLLBACK TO sp1;
SELECT * FROM Employees;

COMMIT;

SELECT * FROM Employees;

SET SQL_SAFE_UPDATES = 1;

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Gaurav	60000.00	10
2	Sneha	75000.00	20
3	Amit	55000.00	30
4	Priya	80000.00	40
5	Rahul	70000.00	50
6	Kiran	65000.00	10
7	Meena	72000.00	20
NULL	NULL	NULL	NULL